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Homework 3: Discrete and Continuous Random Variables

Instructions: Write your answers directly on this pdf (via an editor, iPad, or pen/pencil). The answers should be in the specified place. Students will be responsible for loading their assignments to GradeScope, and identifying what page contains each answer.

The assignment should be uploaded by 11:50pm on the date it is due. There is some slack built into this deadline on GradeScope. Assignments will be marked late if GradeScope marks them late.

If the answers are too hard to read you will lose points (entire questions may be given 0).

Please make sure your name appears at the top of the page.

You may discuss the concepts with your classmates, but write up the answers entirely on your own. **Be sure to show all the work involved in deriving your answers! If you just give a final answer without explanation, you may not receive credit for that question.**

1. Let X be a discrete random variable with probability density function f_X given by:

a	-2	-1	0	1	2
$f_X(a)$	$\frac{3}{10}$	$\frac{1}{10}$	$\frac{1}{5}$	$\frac{1}{10}$	$\frac{3}{10}$

and $f_X(a) = 0$ for all other a .

- (a) Let the random variable Y be defined by $Y = X^2$, i.e. if $X = 2$, then $Y = 4$. Calculate the probability density function of Y .

$$Y = \begin{array}{c|ccc} a & 0 & 1 & 4 \\ \hline f_Y(a) & \frac{1}{5} & \frac{2}{10} & \frac{6}{10} \end{array}$$

(2/10)

$$\frac{2}{10} + \frac{2}{10} + \frac{6}{10} = \frac{10}{10} = 1$$

- (b) Calculate the values of the cumulative distribution function of X and Y in $a = 1$, $a = 3/4$, and $a = \pi - 1$.

$$F(a) = \sum_{a_i \leq a} P(a_i)$$

1) $a = 1$ for X , $\sum f_X(a)$ where $a \leq 1$

$$f_X(-2) + f_X(-1) + f_X(0) + f_X(1) = \frac{3}{10} + \frac{1}{10} + \frac{1}{5} + \frac{1}{10} = \frac{7}{10}$$

for Y , $\sum f_Y(a)$ where $a \leq 1$

$$f_Y(0) + f_Y(1) = \frac{2}{10} + \frac{2}{10} = \frac{4}{10} = \frac{2}{5}$$

2) $a = 3/4$ X : $f_X(-2) + f_X(-1) + f_X(0) \rightarrow \frac{3}{10} + \frac{1}{10} + \frac{1}{5} = \frac{6}{10}$
 Y : $f_Y(0) \rightarrow \frac{2}{10}$

3) $a = 2.14159...$ X : $f_X(-2) + f_X(-1) + f_X(0) + f_X(1) + f_X(2) = \frac{3}{10} + \frac{1}{10} + \frac{2}{10} + \frac{1}{10} + \frac{3}{10} = \frac{10}{10} = 1$

$$Y$$
: $f_Y(0) + f_Y(4) \rightarrow \frac{2}{10} + \frac{2}{10} = \frac{4}{10}$

2. Utah Jazz guard Jordan Clarkson makes 82% of his free throw attempts (and misses 18%). Let's say he will shoot 100 free throws next season. For each of the questions below, please give ALL of the following information: (1) the formula to compute the answer, and (2) the final number for the probability. (You will want to use a calculator or R to compute these.)

(a) What is the probability that Clarkson will miss exactly 16 free throws next season?

Binomial PDF $f_x(k) = \binom{n}{k} p^k (1-p)^{n-k}$

$f_x(84) = \binom{100}{84} .82^{84} (1-.82)^{16}$

$\text{dbinom}(16, 100, .18) = 0.04412407$

(b) What is the probability that Clarkson will make 85 or more free throws next season?

Summed Binomial PDF $\sum_{k=0}^{15} f_x(k) = \sum_{k=0}^{15} \binom{n}{k} p^k (1-p)^{n-k}$

$\sum_{k=0}^{15} \binom{100}{k} .18^k (1-.82)^{100-k}$

R: 1) $y=0$

2) for i in $0:15$) $y = y + \text{dbinom}(i, 100, .18)$

3) $y \approx 0.2020844$

(c) What is the probability that Clarkson will miss all of his free throws next season?

Binomial PDF $f_x(k) = \binom{n}{k} p^k (1-p)^{n-k}$

$f_x(0) = \binom{100}{0} .82^0 (1-.82)^{100}$

$\text{dbinom}(0, 100, .82) = 3.367057 \times 10^{-75}$

3. I bought a cheap toaster. Every time I use it there is a chance that it will catch on fire with probability $p = 0.25$, burn my toast, and destroy itself. But with probability $(1 - p)$ it makes perfect toast!

$$(1 - p) = 1 - 0.25 = 0.75$$

- (a) Suppose we toast slices of bread until a slice is burned. Let X be the number of slices until the first one is burned. What type of random variable is X ?

Geometric Distribution - Discrete Random Variable

• "number of slices until first (Burned Toast) success
With probability $p (= 0.25)$ "

- (b) Suppose we have 20 slices of bread. In R, plot the pdf of the random variable X . The points on your plot should be filled squares and colored red. **Include both your code and your plot in your answer.**

- (c) In R, plot the cdf of X . The type of plot should be a stair step plot. Also, make the line color blue. **Include both your code and your plot in your answer.**

- (d) What is the probability I put at least 7 pieces of bread into my cheap toaster before it destroys itself? **include R code used to get result.**

$$(1 - \text{pgeom}(6, .25)) = 0.13348397$$

• Probability that you have 7 unburned toasts and on the 8th try the toaster blows up with a .25 chance

4. The score of a student on a certain exam is represented by a number between 0 and 1. Suppose that the student passes the exam if this number is at least 0.60. Suppose we model this experiment by a continuous random variable X , the score, whose probability density function is given by

$$f(x) = \begin{cases} 4x & \text{for } 0 \leq x \leq \frac{1}{2} \\ 4 - 4x & \text{for } \frac{1}{2} \leq x \leq 1 \\ 0 & \text{otherwise.} \end{cases}$$

What is the probability that the student fails the exam? Show all your work (i.e., what you are integrating and differentiating, etc.)

$$\text{for } 0 \leq x \leq \frac{1}{2} \quad \int_0^{\frac{1}{2}} 4x \, dx = 2x^2 \rightarrow \left[2\left(\frac{1}{2}\right)^2 - 2(0)^2 \right] \\ \frac{1}{2} - 0 = \boxed{\frac{1}{2}}$$

$$\text{for } \frac{1}{2} \leq x \leq 1 \quad \int_{\frac{1}{2}}^1 4 - 4x \, dx = 4x - 2x^2 \rightarrow \left[(4(1) - 2(1)^2) - (4\left(\frac{1}{2}\right) - 2\left(\frac{1}{2}\right)^2) \right] \\ \left[(4 - 2) - \left(2 - \frac{1}{2}\right) \right] \\ \frac{4}{2} - \frac{3}{2} = \boxed{\frac{1}{2}}$$

$$\text{for } \frac{5}{10} \leq x \leq \frac{6}{10} \quad \int_{\frac{5}{10}}^{\frac{6}{10}} 4 - 4x \, dx = 4x - 2x^2 \rightarrow \left[(4(0.6) - 2(0.6)^2) - (4(0.5) - 2(0.5)^2) \right] \\ 1.08 - 1.5 = \boxed{0.8}$$

$$\text{odds of failing} = \left(\int_0^{\frac{1}{2}} 4x \, dx \right) + \left(\int_{\frac{5}{10}}^{\frac{6}{10}} 4 - 4x \, dx \right) \\ = 0.5 + 0.18 = \boxed{0.68}$$

5. Let X be a continuous random variable that takes values in $[0,1]$, and whose cumulative distribution function F satisfies

$$F(x) = \begin{cases} 0, & \text{if } x < 0 \\ 2x^3 - x^4, & \text{for } 0 \leq x \leq 1 \\ 1, & \text{if } x > 1 \end{cases}$$

- (a) Compute $\Pr(\frac{1}{4} \leq X \leq \frac{3}{4})$

$$\int_{\frac{1}{4}}^{\frac{3}{4}} [2x^3 - x^4] \rightarrow \left[\left(2\left(\frac{3}{4}\right)^3 - \left(\frac{3}{4}\right)^4 \right) - \left(2\left(\frac{1}{4}\right)^3 - \left(\frac{1}{4}\right)^4 \right) \right]$$

$$= 0.5$$

- (b) What is the probability density function of X ?

The PDF can be found by differentiating the CDF by F.T.C.

$$f'(x) = f(x) = \begin{cases} 0, & \text{for } x < 0 \\ 6x^2 - 4x^3, & \text{for } 0 \leq x \leq 1 \\ 0, & \text{for } x > 1 \end{cases}$$